



SAUDI LEDGER PLATFORM

COMPLETE FRONTEND IMPLEMENTATION PROMPT

Version 1.0 | August 30, 2026

Document Type: Technical Implementation Specification
Purpose: Complete frontend build instructions for Saudi Ledger Platform
Target Audience: Frontend Developer / UI/UX Designer
Project Status: Backend 85% Complete | Frontend Needs Complete Overhaul
Estimated Implementation Time: 4-6 weeks (Phase 1), 8-12 weeks (Complete)

TABLE OF CONTENTS

Section	Title	Page
1	Project Overview	1
2	Feature Status Summary	2
3	Technology Stack	4
4	Complete Navigation Structure	5
5	Module Specifications	8
6	Design System	15
7	Component Library	18

8	Integrations Hub	22
9	AI & Automation	25
10	Implementation Priority	28
11	Technical Implementation Notes	29
12	Quality Assurance Checklist	31
13	Deliverables	32

1. PROJECT OVERVIEW

1.1 Background

The Saudi Ledger Platform is a double-entry accounting ERP system designed specifically for the Saudi market. The platform is ZATCA-compliant, VAT-verified, and includes a comprehensive audit trail.

Current Status:

- Backend: ~85% complete
- Frontend: Functional but "temporary"
- Business: ~40% complete (not yet revenue-generating)

1.2 The Challenge

The current frontend is not production-ready. It needs:

1. Complete visual overhaul - Professional, modern design
2. Feature completion - All backend features need polished UIs
3. New modules - Integrations Hub, AI Assistant, PDF generation
4. Mobile responsiveness - Full support across all devices
5. Revenue readiness - Billing, subscriptions, deployment

1.3 Your Mission

Deliver a production-ready frontend that:

- Visually matches Odoo 17 / Linear / Stripe quality
 - Implements all 85% of working backend features
 - Adds critical missing features (PDF, billing, integrations)
 - Works flawlessly in Arabic and English (RTL, Hijri dates)
 - Delights power users with keyboard shortcuts
 - Is fully responsive across all devices
-

2. FEATURE STATUS SUMMARY

2.1 Already Built & Working (Backend)

Category	Features
Ledger	Chart of Accounts, Account Types, Liquidity Classification, VAT Treatment (S/Z/E/O), Input-VAT Blocked Flag
Journal Entries	Double-entry, Draft→Posted→Reversed lifecycle, Period Lock enforcement
Reports	General Ledger, Trial Balance, 14 working reports (P&L, Balance Sheet, Cash Flow, Aging, etc.)
Sales	Invoices, Credit Notes, Debit Notes, Quotations, Customers, AR Aging, Server-side numbering
Purchases	Bills, Purchase Orders, Vendors, AP Aging, Price Variance Recording
Banking	Statement Import (CSV), Automatic Categorisation, Transfer Detection, Reconciliation
Tax	VAT Return (box-structured, bilingual), VAT Reconciliation, ZATCA Phase 1 & 2 (QR, UBL 2.1 XML, ECDSA signing)

Controls	Multi-tenancy, 4 Roles (Admin, Accountant, Bookkeeper, Viewer), Permissions Matrix, Approval Workflow, Audit Trail
Automation	Receipt Capture (QR decode + OCR), Recurring Invoices/Bills (drafts only), Findings Engine
Interface	Bilingual UI (Arabic/English), RTL groundwork, Hijri dates, Pagination

2.2 🛑 Built but Blocked / Pending

- ZATCA Production Submission (needs Saudi VAT registration)
- AI Explanations & Grounded CFO (needs Groq Enterprise)
- Live Bank Feeds (needs SAMA-licensed provider)

2.3 🛠️ In Progress / Design Ready

- Sub-accounts / hierarchy
- Account numbering scheme
- Cost centres / project tracking
- Customer / Vendor detail pages
- Password reset (needs email provider)

2.4 🛑 Not Built – Critical Gaps

Priority	Feature	Why Critical
1	PDF Document Generation	Invoices can't be printed or sent
2	Visual Design / Brand Identity	Current look is temporary
3	Billing / Subscriptions	Can't take payment
4	Deployment	No hosted instance

5	Email Delivery	Invitations, notifications reach nobody
6	Mobile Responsive	Not assessed
7	Inventory / COGS	Blocks trading businesses
8	Multi-Currency	Blocks import businesses
9	Fixed Asset Depreciation	Feeds Zakat base
10	Payroll / GOSI	Salary categories exist, no payroll runs
11	Zakat Calculation	Blocked on tax advice
12	Integrations	No POS, ERP, bank feeds

3. TECHNOLOGY STACK

3.1 Frontend Framework

text

Framework: React 19 + TypeScript

Build Tool: Vite

Routing: Wouter (or React Router)

State: TanStack Query (React Query)

Styling: Tailwind CSS v4 + shadcn/ui

Forms: react-hook-form + Zod

Charts: Recharts or Chart.js

PDF: @react-pdf/renderer or jsPDF

Monorepo: pnpm workspaces (apps/web, apps/api)

3.2 Development Environment

text

Node.js: v18+

Package Manager: pnpm

TypeScript: v5+

Browser Support: Modern browsers (Chrome, Firefox, Safari, Edge)

3.3 Key Dependencies

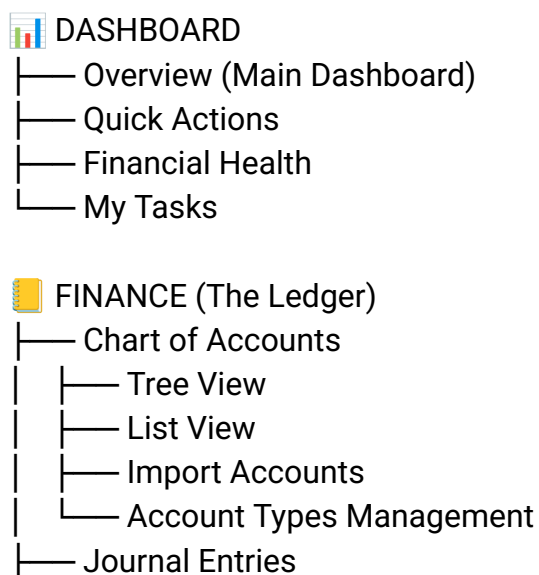
json

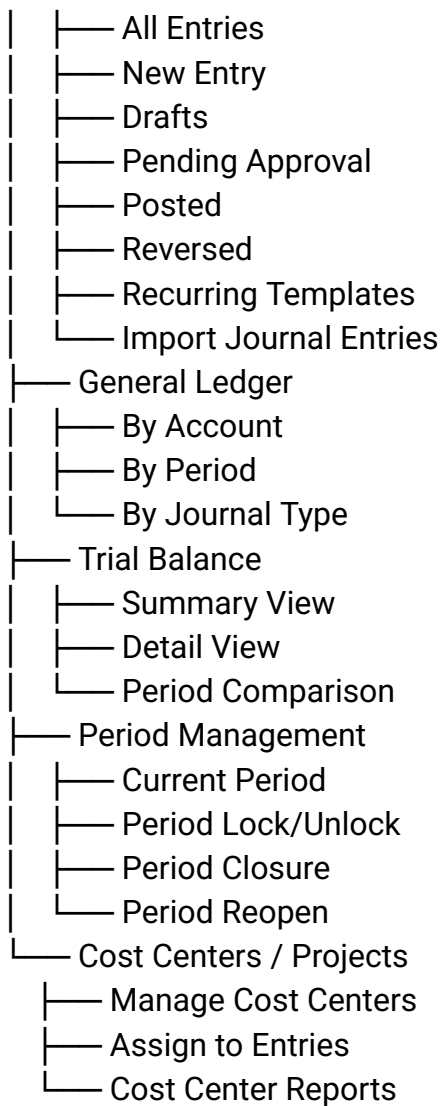
```
{  
  "react": "^19.0.0",  
  "react-dom": "^19.0.0",  
  "typescript": "^5.0.0",  
  "@tanstack/react-query": "^5.0.0",  
  "tailwindcss": "^4.0.0",  
  "@hookform/resolvers": "^3.0.0",  
  "zod": "^3.0.0",  
  "recharts": "^2.0.0",  
  "@react-pdf/renderer": "^3.0.0"  
}
```

4. COMPLETE NAVIGATION STRUCTURE

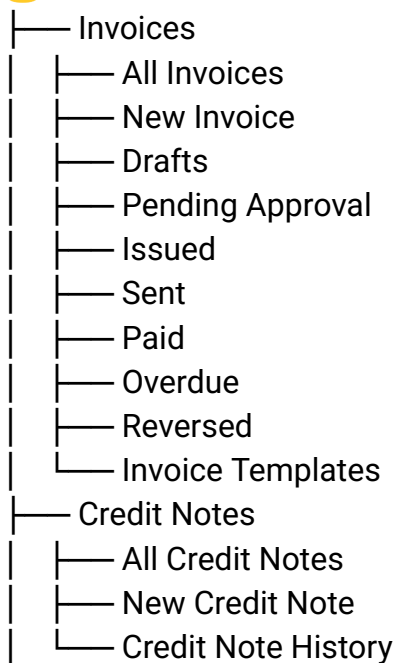
4.1 Sidebar Menu – Full Hierarchy

text





SALES



- └─ Debit Notes
 - └─ All Debit Notes
 - └─ New Debit Note
- └─ Quotations
 - └─ All Quotations
 - └─ New Quotation
 - └─ Converted to Invoice
 - └─ Expired
 - └─ Quotation Templates
- └─ Customers
 - └─ All Customers
 - └─ New Customer
 - └─ Customer Detail
 - └─ Customer Statements
 - └─ Customer Aging
 - └─ Customer Groups
- └─ AR Aging
 - └─ Summary
 - └─ Detail
 - └─ Aging Report
- └─ Sales Reports
 - └─ Sales by Customer
 - └─ Sales by Product
 - └─ Sales by Region
 - └─ Sales Trends



PURCHASES

- └─ Bills
 - └─ All Bills
 - └─ New Bill
 - └─ Drafts
 - └─ Pending Approval
 - └─ Paid
 - └─ Overdue
 - └─ Reversed
- └─ Purchase Orders
 - └─ All Purchase Orders
 - └─ New Purchase Order
 - └─ Converted to Bill
 - └─ PO Templates
- └─ Vendors
 - └─ All Vendors

- | — New Vendor
- | — Vendor Detail
- | — Vendor Statements
- | — Vendor Aging
- | — AP Aging
- | — Summary
- | — Detail
- | — Aging Report
- | — Purchase Reports
- | — Purchases by Vendor
- | — Purchases by Product
- | — Purchase Trends



BANKING

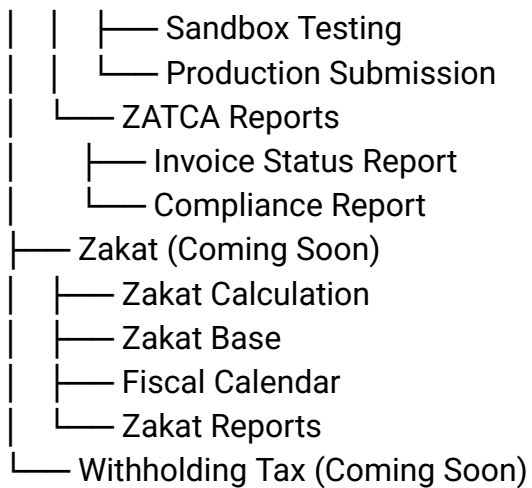
- | — Bank Accounts
- | — All Accounts
- | — New Bank Account
- | — Account Detail
- | — Account Reconciliation
- | — Bank Statements
- | — All Statements
- | — Import Statement
- | — Statement Review Queue
- | — Categorisation
- | — Bulk Accept
- | — Statement History
- | — Transfers
- | — All Transfers
- | — New Transfer
- | — Internal Transfers
- | — External Transfers
- | — Reconciliation
- | — Unreconciled Transactions
- | — Reconciliation Suggestions
- | — Manual Matching
- | — Reconciliation History
- | — Live Bank Feeds (Coming Soon)
- | — Connect Bank
- | — Connected Banks
- | — Sync Schedule
- | — Transaction Feed
- | — Banking Reports

- Cash Flow
- Bank Balance History
- Transfer Reports

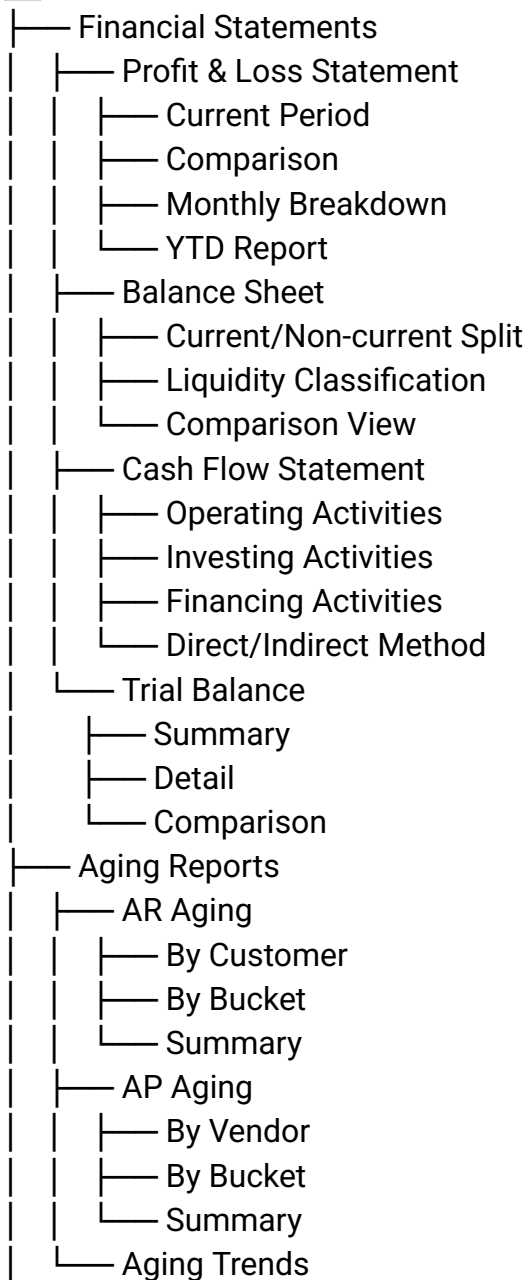


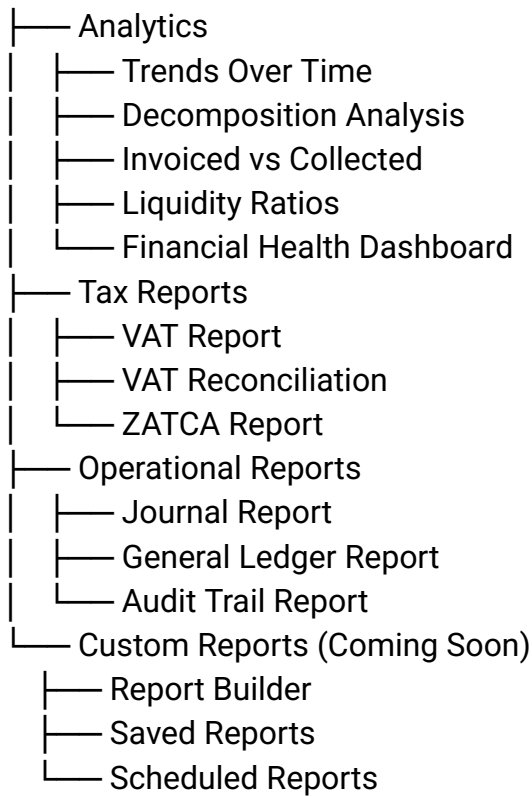
TAX

- VAT
 - VAT Return
 - Generate Return
 - VAT Boxes
 - Bilingual Display
 - VAT Reconciliation View
 - Submit VAT Return
 - VAT Reconciliation
 - Documents vs Bank
 - Blocked Input VAT
 - VAT Treatment Verification
 - VAT Reports
 - Sales VAT
 - Purchase VAT
 - Net VAT Summary
- ZATCA E-Invoicing
 - Phase 1 (Hash chain, QR)
 - QR Code Generation
 - Hash Chain Management
 - Phase 2 (UBL 2.1 XML)
 - XML Generation
 - XML Preview
 - XML Validation
 - XML Download
 - Cryptographic Signing
 - ECDSA secp256k1
 - XAdES Signature
 - Certificate Management
 - Onboarding
 - Company Registration
 - CSR Generation
 - CSID Request
 - Production CSID
 - Onboarding Status
 - ZATCA Compliance Dashboard
 - Submission History
 - Compliance Documents

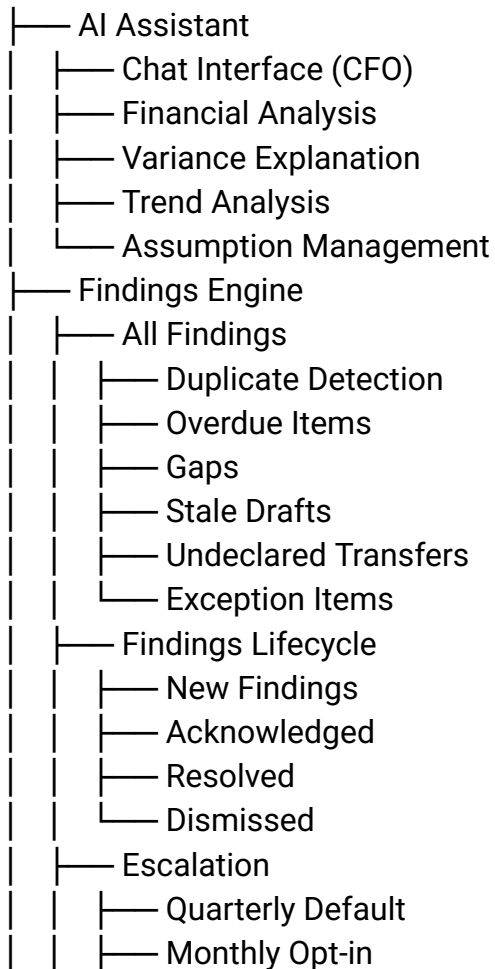


REPORTS





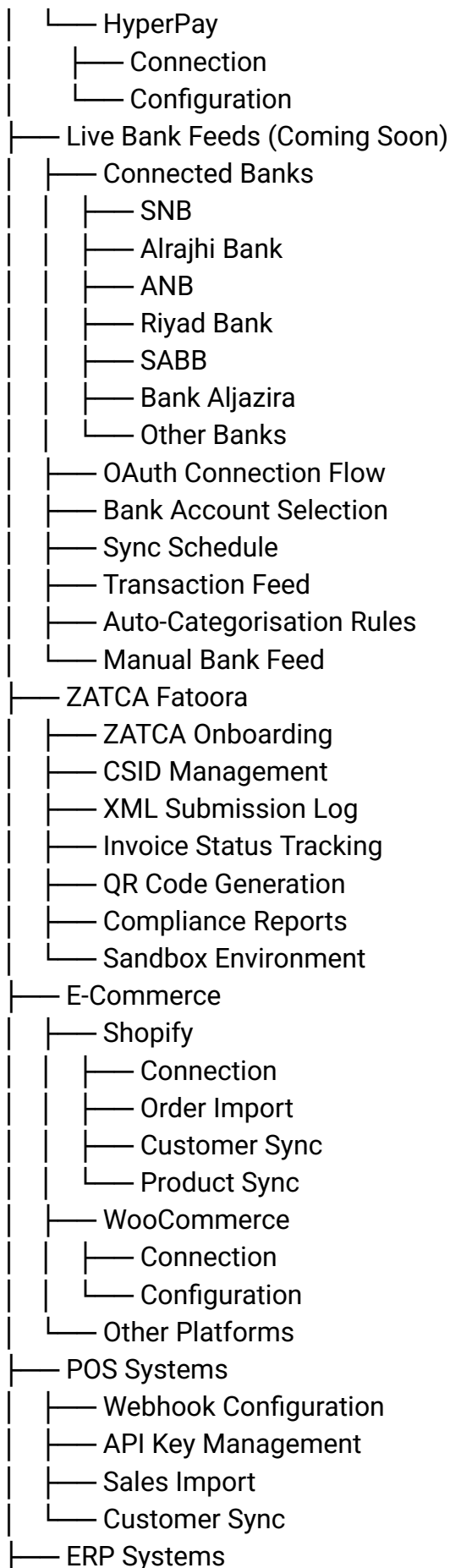
AI & AUTOMATION

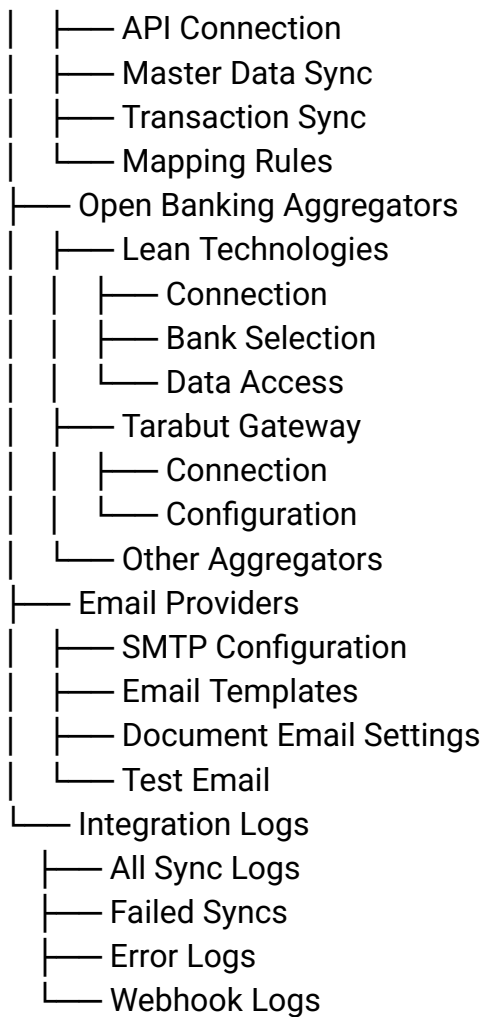


- └─ Escalation Rules
 - └─ AI Explanations (Coming Soon)
- └─ Receipt Capture
 - └─ Camera Capture
 - └─ ZATCA QR Decode
 - └─ OCR Fallback
 - └─ Supplier Auto-Match
 - └─ Review & Post
 - └─ Image Storage
- └─ Recurring Transactions
 - └─ Recurring Invoices
 - └─ Create Rule
 - └─ Draft Generation
 - └─ Failed Run Recording
 - └─ Recurring Bills
 - └─ Create Rule
 - └─ Draft Generation
 - └─ Recurring Journal Entries (Coming Soon)
 - └─ Run History
 - └─ Successful Runs
 - └─ Failed Runs
- └─ Vision Model (Coming Soon)
 - └─ Receipt Recognition
 - └─ Invoice Recognition
 - └─ Document Classification

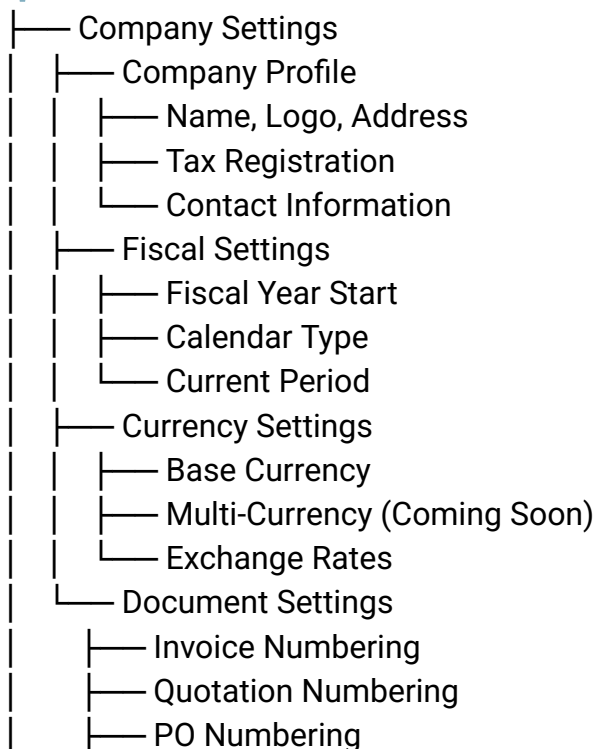
INTEGRATIONS (NEW)

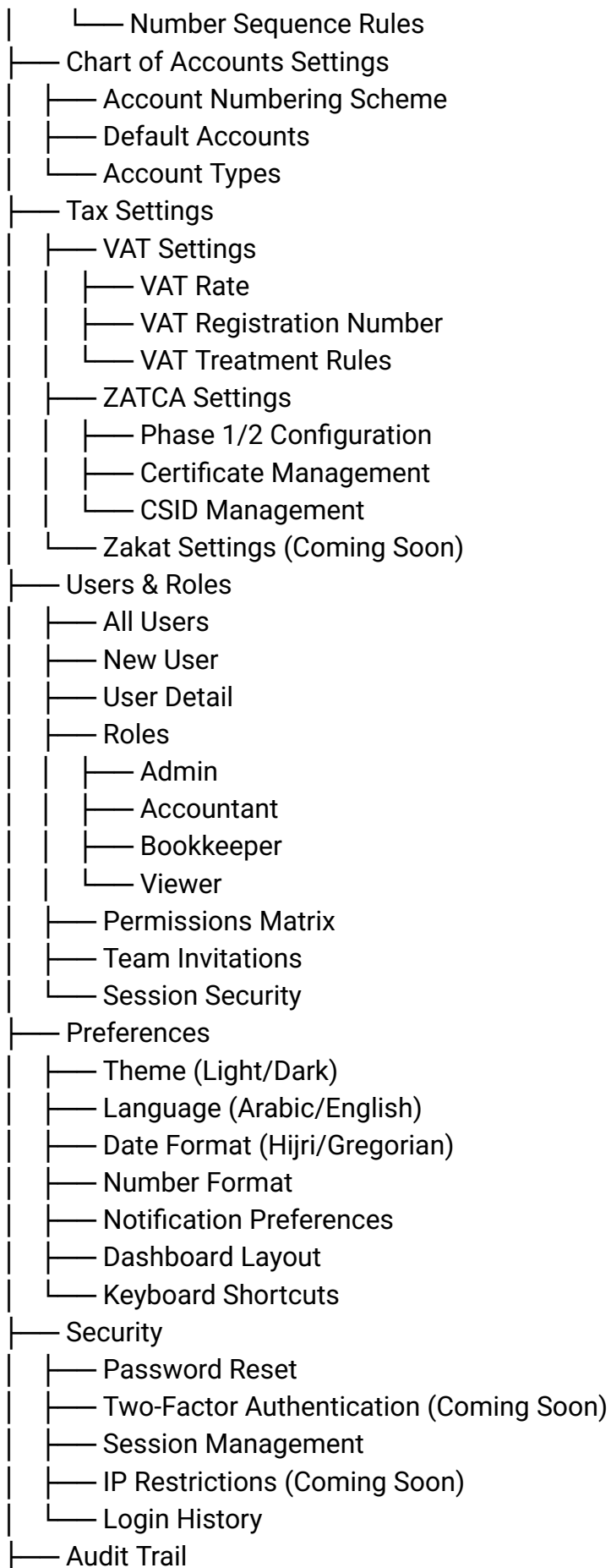
- └─ Integration Dashboard
 - └─ Connection Status
 - └─ Quick Actions
 - └─ Category Groups
- └─ Payment Gateways
 - └─ MyFatoorah
 - └─ Connection (Sandbox/Live)
 - └─ API Keys Management
 - └─ Webhook Configuration
 - └─ Auto-Reconcile Settings
 - └─ Payment History
 - └─ Test Connection
 - └─ PayTabs
 - └─ Connection
 - └─ Configuration

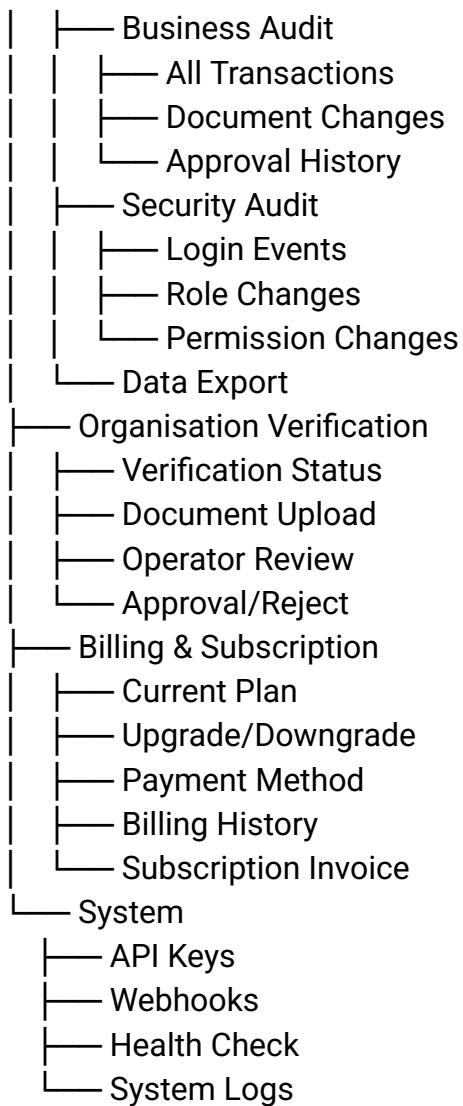




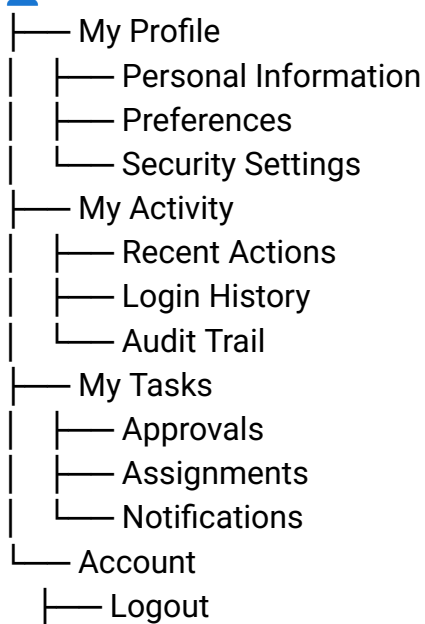
SETTINGS







USER PROFILE



5. MODULE SPECIFICATIONS

5.1 Dashboard

Purpose: Real-time financial overview for executives and managers.

Layout:

text

```
+-----+
| KPI ROW (5 cards)                |
| [Total Assets] [A/R] [A/P] [Cash Flow] [Net P&L] |
+-----+
| LEFT (60%)          | RIGHT (40%)      |
| Revenue/Expense Line Chart | Aging Summary      |
| + P&L Trend Chart    | + Top Customers    |
+-----+-----+
| Recent Transactions Table (10 rows)      |
+-----+
| Accounting Equation: Assets = Liabilities + Eq |
+-----+
```

KPI Cards:

- Large, bold numbers with trend indicators (↑ green / ↓ red)
- Currency formatting with thousand separators
- Click through to detailed reports
- Hover shows additional context

Charts (Recharts/Chart.js):

- Line chart: Revenue vs Expenses (12-month rolling)
- Bar chart: Monthly P&L with profit margin overlay
- Donut chart: Expense breakdown by category

Key Lists:

- Overdue A/R: Show invoices past due by aging buckets
- Recent Transactions: Last 10 entries with status badges

Accounting Equation:

- Prominently display: Assets = Liabilities + Equity
- Real-time balance status (✓ Balanced / ✗ Unbalanced)

5.2 Journal Entries

Purpose: The heart of the accounting system.

List View Columns:

| # | Date | Journal | Reference | Description | Debit | Credit | Status | Actions |

Status Badges:

- Draft (yellow)
- Pending Approval (orange)
- Posted (green)
- Reversed (red)
- Cancelled (grey)

Create/Edit Form – CRITICAL FOR EFFICIENCY:

text

```
+-----+
| JOURNAL ENTRY |
| Date: [ ] Journal: [▼] Reference: [ ] |
+-----+
| # | Account | Description | Debit | Credit |
|---+-----+-----+-----+-----|
| 1 | [ ] | [ ] | [ ] | [ ] |
| 2 | [ ] | [ ] | [ ] | [ ] |
| 3 | [ ] | [ ] | [ ] | [ ] |
| + Add Line |
+-----+
| Total Debit: [1,000.00] Total Credit: [1,000.00] |
| [✓] Balanced |
+-----+
| [Save Draft] [Submit] [Cancel] |
+-----+
```

Key Interaction Requirements:

- Tab key navigation (field to field without mouse)

- Account auto-complete (type code or name, suggest matches)
- Real-time balancing (show total debit vs. credit, highlight if unbalanced)
- Quick duplicate (copy existing entry with new date)
- Keyboard shortcuts: Ctrl+S (Save), Ctrl+Enter (Submit), Esc (Cancel)

5.3 Invoices (A/R)

Purpose: Manage customer invoices and collections.

Invoice Form:

text

```

+-----+
| INVOICE # [AUTO-GENERATED] |
| Customer: [Search/Select]  Date: [____] |
| Due Date: [____]          Currency: [SAR ▼] |
+-----+
| # | Product/Service | Qty | Unit Price | Tax | Total |
|---+-----+---+-----+---+-----|
| 1 | [_____] | [.] | [_____] | [%] | [_____] |
| 2 | [_____] | [.] | [_____] | [%] | [_____] |
| + Add Line |
+-----+
| Subtotal: [_____] VAT: [_____] Total: [1,000.00] |
| ZATCA QR: [QR Code Preview] |
+-----+
| Status: [Draft ▼] |
| [Save Draft] [Submit for Approval] [Issue] |
| [Send Email] [Record Payment] [Print PDF] |
+-----+

```

Critical Features:

- ZATCA compliance: QR code generation, XML/JSON formatting
- Invoice numbering: Sequential, no gaps (server-generated)
- Payment tracking: Partial payments, remaining balance
- Email integration: Send invoice directly to customer
- Print/PDF: Properly formatted with company letterhead
- Status workflow: Draft → Sent → Partial → Paid → Overdue

5.4 Customer/Vendor Detail Pages

Layout:

text

+-----+	
Customer: [Company Name] [Edit]	
Status: Active Credit Limit: 50,000 SAR	
+-----+	
Summary Cards	
[Total Sales] [Outstanding] [Avg Payment]	
+-----+	
Tabs:	
[Overview] [Invoices] [Payments] [Quotes] [Docs]	
+-----+	
[Overview Content]	
- Contact Information	
- Tax Details (VAT, CR)	
- Activity Timeline	
- Recent Transactions	
+-----+	

5.5 Reports Center

Report Interface:

text

+-----+	
REPORT: Profit & Loss Statement	
Period: [Q4 2026 ▼] Compare: [Q3 2026 ▼]	
Format: [PDF ▼] Include: [All Accounts ▼]	
+-----+	
[GENERATE REPORT] [EXPORT] [EMAIL] [SCHEDULE]	
+-----+	
(Report preview with proper formatting)	
+-----+	

Export Requirements:

- PDF: Properly formatted for tax authorities
- Excel (XLSX): With formulas intact
- CSV: For data import into other systems

6. DESIGN SYSTEM

6.1 Color Palette

Light Mode:

Usage	Hex	Description
Primary	#1E40AF	Deep royal blue - trust & authority
Primary Light	#3B82F6	Bright blue - interactions
Primary Dark	#1E3A8A	Dark blue - emphasis
Secondary	#D4A843	Saffron gold - premium feel
Success	#059669	Emerald - paid, balanced, positive
Warning	#D97706	Amber - pending, draft, caution
Danger	#DC2626	Red - overdue, error, negative
Info	#2563EB	Blue - information
Background	#F8FAFC	Page background
Surface	#FFFFFF	Card/panel background
Surface Hover	#F1F5F9	Hover state
Border	#E2E8F0	Borders
Text Primary	#0F172A	Main text
Text Secondary	#475569	Secondary text
Text Muted	#94A3B8	Muted text

Dark Mode:

Usage	Hex	Description
Background	#0F172A	Page background
Surface	#1E293B	Card/panel background
Surface Hover	#334155	Hover state
Border	#334155	Borders
Text Primary	#F1F5F9	Main text
Text Secondary	#94A3B8	Secondary text

6.2 Typography

Font Family:

- Arabic: 'Noto Sans Arabic', 'Amiri', sans-serif
- English: 'Inter', 'SF Pro Display', -apple-system, sans-serif
- Numbers: 'JetBrains Mono', 'SF Mono', monospace

Scale:

Size	rem	px	Usage
xs	0.75rem	12px	Captions, metadata
sm	0.875rem	14px	Body text, table cells
base	1rem	16px	Default
lg	1.125rem	18px	Card titles
xl	1.25rem	20px	Section headers
2xl	1.5rem	24px	Page titles

3xl	1.875rem	30px	Dashboard KPI
4xl	2.25rem	36px	Hero numbers

Number Formatting:

- Thousand separators: comma (1,000.00)
- Decimal places: 2 (configurable)
- Currency symbol: SAR
- Negative: parentheses (1,000.00)
- Alignment: Right-aligned for numbers

6.3 Shadow System

Size	CSS
sm	0 1px 2px rgba(0,0,0,0.05)
md	0 4px 6px rgba(0,0,0,0.07)
lg	0 10px 15px rgba(0,0,0,0.1)
xl	0 20px 25px rgba(0,0,0,0.12)

6.4 Spacing System

Based on 4px grid:

- 0.25rem = 4px
- 0.5rem = 8px
- 0.75rem = 12px
- 1rem = 16px
- 1.5rem = 24px
- 2rem = 32px
- 3rem = 48px
- 4rem = 64px

6.5 Border Radius

- sm: 0.25rem (4px)
 - md: 0.375rem (6px)
 - lg: 0.5rem (8px)
 - xl: 0.75rem (12px)
 - full: 9999px
-

7. COMPONENT LIBRARY

7.1 Data Table Component

typescript

```
interface DataTableProps<T> {  
  data: T[];  
  columns: ColumnDef<T>[];  
  pageSize?: number;  
  showPageSize?: boolean;  
  showSearch?: boolean;  
  showFilters?: boolean;  
  showColumnVisibility?: boolean;  
  virtualScroll?: boolean;    // For > 200 rows  
  rowSelection?: boolean;  
  actions?: (row: T) => React.ReactNode;  
  onRowClick?: (row: T) => void;  
  loading?: boolean;  
}
```

Features:

- Fixed header when scrolling
- Column sorting (click header)
- Column resizing (drag to adjust)
- Column visibility toggle
- Row selection with bulk actions
- Zebra striping (alternating row colors)
- Virtual scrolling for large datasets

7.2 Financial Input Component

typescript

```
interface FinancialInputProps {  
  value: number;  
  onChange: (value: number) => void;  
  currency?: 'SAR' | 'USD' | 'EUR';  
  locale?: 'en' | 'ar';  
  negativeAllowed?: boolean;  
  placeholder?: string;  
  className?: string;  
}
```

Features:

- Right-aligned
- Thousand separators
- Decimal places (2)
- Negative in parentheses
- Arabic-Indic digit support
- Keyboard-friendly

7.3 Status Badge Component

typescript

```
type StatusType =  
  | 'draft' | 'pending' | 'approved' | 'issued'  
  | 'sent' | 'paid' | 'partial' | 'overdue'  
  | 'cancelled' | 'reversed' | 'posted' | 'verified';
```

```
interface StatusBadgeProps {  
  status: StatusType;  
  label?: string;  
  size?: 'sm' | 'md' | 'lg';  
}
```

Styles:

- Draft: Yellow
- Pending: Orange
- Approved: Blue
- Paid: Green
- Overdue: Red
- Cancelled: Grey

7.4 Dual Date Picker Component

typescript

```
interface DualDatePickerProps {  
  value: Date;  
  onChange: (date: Date, source: 'gregorian' | 'hijri') => void;  
  locale: 'en' | 'ar';  
  displayMode: 'both' | 'gregorian' | 'hijri';  
  calendarType: 'UmmAlQura' | 'Gregorian';  
}
```

7.5 KPI Card Component

typescript

```
interface KpiCardProps {  
  title: string;  
  value: number;  
  trend?: number;           // Percentage change  
  currency?: string;  
  onClick?: () => void;      // Drill down  
  loading?: boolean;  
}
```

7.6 Button Component

typescript

```
interface ButtonProps {  
  variant: 'primary' | 'secondary' | 'danger' | 'ghost' | 'link';  
  size: 'sm' | 'md' | 'lg';  
  icon?: React.ReactNode;  
  loading?: boolean;  
  disabled?: boolean;  
  onClick?: () => void;  
  children: React.ReactNode;  
}
```

7.7 Modal Component

Features:

- Rounded corners, consistent shadow
- Escape key to close
- Click outside to close (for non-critical modals)
- Proper dark mode support
- Focus trapping

7.8 Toast/Notification Component

Features:

- Left-border style (no heavy colored backgrounds)
 - Auto-dismiss after 5 seconds
 - Stack multiple notifications
 - Types: Success (green), Warning (yellow), Error (red), Info (blue)
-

8. INTEGRATIONS HUB

8.1 Overview Dashboard

```
text
+-----+
| INTEGRATIONS DASHBOARD |
+-----+
| Connected: 5 | Available: 8 | Errors: 2 |
+-----+
| Quick Actions: [Connect New] [Sync All] [Logs] |
+-----+
| Category: Payment Gateways |
| [MyFatoorah] [PayTabs] [HyperPay] |
| [Connected✅] [Available] [Available] |
+-----+
| Category: Bank Feeds |
| [SNB] [Alrajhi] [ANB] [Riyad Bank] |
| [Error⚠️] [Connected✅] [Available] [Available] |
+-----+
| Category: E-commerce |
| [Shopify] [WooCommerce] |
```

| [Available] [Available] |
+-----+
| Category: E-Invoicing |
| [ZATCA Onboarding] [Fatoora Sandbox] |
| [Connected ] [Available] |
+-----+

8.2 Payment Gateway Configuration (MyFatoorah)

text

+-----+
| MyFatoorah Configuration |
+-----+
| Environment: [Live ▼] |
| API Key: [.....] [Show] |
| Secret Key: [.....] [Show] |
+-----+
| Webhook URL: |
| https://api.saudi-ledger.com/webhooks/fatoorah |
| [Copy] |
+-----+
| Automation Rules: |
| ☒ Auto-reconcile payment to invoice |
| ☒ Send confirmation email |
| ☒ Mark invoice as "Paid" instantly |
| ☐ Auto-create customer if not exists |
+-----+
| [Test Connection] [Save Changes] [Cancel] |
+-----+

8.3 Live Bank Feeds Connection Flow

text

+-----+
| CONNECT BANK ACCOUNT |
+-----+
| Select Bank: |
| [Saudi National Bank (SNB)] |
| [Alrajhi Bank] |

[Arab National Bank (ANB)]	
[Riyad Bank]	
[SABB]	
[Other Bank]	
+-----+	
Permissions Requested:	
✓ Read account balance	
✓ Read transaction history	
✓ Read beneficiary information	
✗ Transfer funds (not requested)	
+-----+	
Data Access Period: 90 days	
+-----+	
[Proceed to Bank] [Cancel]	
+-----+	

8.4 ZATCA Fatoora Dashboard

text

+-----+				
ZATCA E-INVOICING				
+-----+				
Phase 1 (Hash Chain, QR) ✓				
Phase 2 (UBL 2.1 XML) ✓				
Cryptographic Signing ✓				
+-----+				
Onboarding Status:				
+-----+				
✓ Company Registration				
✓ CSR Generated				
⌚ CSID Request Submitted				
☐ CSID Received				
☐ Production Ready				
+-----+				
Invoice Submission Log:				
+-----+				
Date Invoice # Status UUID Action				
+-----+				
30/08 INV-001 Cleared 7a2... View XML				
29/08 INV-002 Rejected — Resubmit				

+-----+

8.5 Integration Logs

text

+-----+

| INTEGRATION LOGS |

+-----+

| Filter: [All Integrations ▼] [Last 7 days ▼] |

| [Search logs...] |

+-----+

| Time | Integration | Type | Status | Details |

|-----|

| 10:45 | MyFatoorah | Sync |  Success | 5 payments processed |

| 09:30 | SNB | Sync |  Partial | 3 new transactions |

| 08:15 | Shopify | Sync |  Error | Rate limit exceeded |

+-----+

| [View Details] [Export Logs] |

+-----+

9. AI & AUTOMATION

9.1 AI Assistant

text

+-----+

| AI FINANCIAL ASSISTANT |

+-----+

| [New Chat] [History] [Analytics] |

+-----+

| |-----| |

| | Chat Interface | |

| | |

| | User: What were our top expenses Q3? | |

| | |

| | AI: Based on your Q3 2026 data: | |

| | • Salaries: SAR 450,000 (42%) | |

| | • Rent: SAR 120,000 (11%) | |

[\[Change to Monthly\]](#) [\[View Escalation History\]](#)

+-----+

9.3 Receipt Capture

text

+-----+

| RECEIPT CAPTURE |

+-----+

|  Capture] [Upload Image] [Scan QR] |

+-----+

| Recent Captures: |

| _____ |

| Date | Vendor | Amount | Status | Source |

|_____| |_____| |_____| |_____| |_____|

| Today | Jarir | SAR 450| Posted | QR Decoded |

| Today | Panda | SAR 89 | Review | OCR'd |

+-----+

| Review Queue: (2 items) |

| _____ |

|  Receipt from Panda (SAR 89) | |

| Supplier: Auto-matched ✓ | |

| Vendor: Panda (VAT: 3012345678) | |

| Date: 30/08/2026 | |

| Items: Groceries | |

| VAT: SAR 11.61 (15%) | |

| [Accept] [Edit] [Reject] | |

| _____ |

+-----+

| Image Retention: |

| All receipts stored with provenance: |

| • QR Decoded (✓) |

| • OCR'd (✓) |

| • Typed () |

+-----+

9.4 Recurring Transactions

text

4	Customer / Vendor Detail Pages	Activity timeline, statements, aging
5	Dark / Light Theme Toggle	Persistent preference, system detection
6	Mobile Responsive	All breakpoints working (mobile, tablet, desktop)

Phase 2 – High Value (Soon After Launch) – Week 5-8

Priority	Module	Deliverables
7	Integrations Hub	Payment gateways, ZATCA onboarding UI
8	AI Assistant (Findings Engine)	Duplicate detection, overdue, gaps UI
9	Receipt Capture	QR decode, OCR, review queue
10	Recurring Transactions	Rules management, run history

Phase 3 – Future Enhancements – Week 9-12

Priority	Module	Deliverables
11	Inventory Management	COGS, stock tracking
12	Multi-Currency	Currency selection, exchange rates
13	Payroll (GOSI)	Payroll runs, GOSI calculations

14	Fixed Asset Depreciation	Asset management, depreciation UI
15	Vision Model	Receipt recognition

11. TECHNICAL IMPLEMENTATION NOTES

11.1 Folder Structure

text

```

apps/web/
├── src/
│   ├── components/
│   │   ├── ui/           # Base components (Button, Input, Table, etc.)
│   │   ├── forms/        # Form components (JournalEntryForm, etc.)
│   │   ├── layouts/      # Layout components (AppLayout, AuthLayout)
│   │   └── modules/      # Module-specific components
│   ├── hooks/
│   │   ├── useKeyboardShortcuts.ts
│   │   ├── useTheme.ts
│   │   ├── useLanguage.ts
│   │   ├── useBreakpoint.ts
│   │   └── useErrorHandler.ts
│   ├── lib/
│   │   ├── api/
│   │   │   ├── client.ts
│   │   │   ├── endpoints/
│   │   │   └── hooks/
│   │   ├── constants/
│   │   ├── types/
│   │   └── utils/
│   └── pages/
│       ├── dashboard/
│       ├── finance/
│       ├── sales/
│       ├── purchases/
│       ├── banking/
│       └── tax/

```

```

| | | — reports/
| | | — ai/
| | | — integrations/
| | | — settings/
| | | — profile/
| | — routes/
| |   — index.tsx
| | — contexts/
| |   — ThemeContext.tsx
| |   — LanguageContext.tsx
| |   — AuthContext.tsx
| | — styles/
| |   — tailwind.css
| |   — dark-mode.css
| | — App.tsx
| | — main.tsx
| — public/
| — index.html
| — package.json
| — tailwind.config.js
| — vite.config.ts
|
| — tsconfig.json

```

11.2 API Integration Pattern

typescript

```

// Example API hook with proper typing
export function useInvoices(filters: InvoiceFilters) {
  return useQuery({
    queryKey: ['invoices', filters],
    queryFn: async () => {
      const response = await apiFetch<ApiResponse<Invoice[]>>('/invoices', {
        params: filters,
      });

      if (!response.success) {
        throw new ApiError(response.error);
      }

      return response.data;
    },
  },

```

```
    staleTime: 30000,  
  });  
}
```

11.3 Error Handling Pattern

typescript

```
// Global error handling  
export function useErrorHandler() {  
  const { toast } = useToast();  
  
  return {  
    handleError: (error: unknown) => {  
      if (error instanceof ApiError) {  
        toast({  
          title: error.code,  
          description: error.message,  
          variant: 'destructive',  
        });  
  
        // Field-level errors  
        if (error.details) {  
          Object.entries(error.details).forEach(([field, message]) => {  
            form.setError(field, { message });  
          });  
        }  
      } else {  
        toast({  
          title: 'Error',  
          description: 'An unexpected error occurred',  
          variant: 'destructive',  
        });  
      }  
    },  
  });  
}
```

11.4 Keyboard Shortcuts Hook

typescript

```

export function useKeyboardShortcuts(shortcuts: Record<string, () => void>) {
  useEffect(() => {
    const handler = (e: KeyboardEvent) => {
      const key = `${e.ctrlKey ? 'Ctrl+' : ''}${e.key}`;

      if (shortcuts[key]) {
        e.preventDefault();
        shortcuts[key]();
      }
    };

    window.addEventListener('keydown', handler);
    return () => window.removeEventListener('keydown', handler);
  }, [shortcuts]);
}

// Usage
useKeyboardShortcuts({
  'Ctrl+S': () => saveDraft(),
  'Ctrl+Enter': () => submitEntry(),
  'Esc': () => closeDialog(),
  'Alt+N': () => addLineItem(),
});

```

11.5 Keyboard Shortcuts Reference

Shortcut	Action	Context
⌘K / Ctrl+K	Global search	Anywhere
⌘N / Ctrl+N	New item	Module-specific
⌘S / Ctrl+S	Save form	Forms
⌘Enter	Submit form	Forms
Esc	Close modal	Modals
Tab	Next field	Forms

Shift+Tab	Previous field	Forms
Alt+N	New line item	Journal/Invoice
Del	Delete selected line	Journal/Invoice

12. QUALITY ASSURANCE CHECKLIST

12.1 Accounting Logic

- ☐ All journal entries balance (debits = credits) – UI shows balance status
- ☐ Accounting equation holds – visible on dashboard
- ☐ Invoice totals match line items (including VAT)
- ☐ ZATCA QR codes are valid and previewable
- ☐ Period lock enforcement – UI refuses to post in locked periods
- ☐ All 14 reports display correctly with comparison options

12.2 UI/UX

- ☐ All pages load within 3 seconds
- ☐ Keyboard shortcuts work in every form
- ☐ Forms validate on blur (not just on submit)
- ☐ Error messages are clear and field-specific
- ☐ Dark mode works across all pages
- ☐ Arabic UI is fully mirrored and readable
- ☐ Hijri calendar support works correctly
- ☐ Mobile layout is usable for critical tasks

12.3 Performance

- ☐ Virtual scrolling for > 200 rows
- ☐ Debounced search (300ms)
- ☐ Lazy loading for reports
- ☐ React.memo for expensive components
- ☐ Proper query caching (TanStack Query)

12.4 Security

- ❑ Role-based permissions enforced
- ❑ No sensitive data in client-side storage
- ❑ API requests use secure tokens
- ❑ Audit trail logs all actions

12.5 Integration

- ❑ Payment gateways connect/disconnect
 - ❑ Bank feeds OAuth flow works
 - ❑ ZATCA onboarding flow works
 - ❑ Integration logs show sync activity
 - ❑ Webhook configuration works
-

13. DELIVERABLES

13.1 Code

- ❑ Complete source code for the entire frontend
- ❑ All components, pages, hooks, utilities, styles
- ❑ API integration layer (all endpoints)
- ❑ Type definitions using API schemas
- ❑ Keyboard shortcuts system
- ❑ Theme configuration (light/dark, RTL)

13.2 Documentation

- ❑ Component library documentation (Storybook if possible)
- ❑ API integration guide
- ❑ Keyboard shortcuts reference
- ❑ Theme customization guide
- ❑ Deployment instructions
- ❑ Testing guide

13.3 Design Assets

- ❑ Complete color palette (light and dark)
- ❑ Typography system (font sizes, weights, spacing)
- ❑ Component library (Storybook or design system)

- UI mockups for all screens
-

14. FINAL NOTES

14.1 Reference Existing Codebase

You have a working but temporary frontend. Enhance, don't reinvent. Use the existing code structure where possible.

14.2 Use shadcn/ui Components

Use shadcn/ui components wherever possible, but customise them to match the financial look and feel.

14.3 Accessibility

- WCAG 2.1 AA compliance minimum
- Keyboard navigation for all interactive elements
- Screen reader support (ARIA labels)
- Color contrast ratio: at least 4.5:1 for text
- Focus indicators visible on all interactive elements

14.4 Testing

- Test with both Arabic and English
- Some users will use English, others Arabic
- Mock the integrations UI even if backend isn't ready
- Show the expected flow for "Coming Soon" features

14.5 Documentation

- Document any assumptions
 - Clearly mark features that are "Coming Soon"
 - Provide inline comments for complex logic
-

15. ACTION

Start building. Use this prompt as your single source of truth.

If you need to prioritise, follow the Phase 1 list first.

Feel free to ask for clarification on any specific module.

Goal: A demo-ready, revenue-ready frontend that rivals Odoo and Qoyod, fully leveraging the 85% complete backend.

Good luck – you've got this! 💪



PDF GENERATION INSTRUCTIONS

Option 1: Convert in Your Browser

1. Copy the entire prompt (it's all above)
2. Paste into Google Docs or Microsoft Word
3. File → Download → PDF Document (.pdf)

Option 2: Use Browser Print

1. Paste into any text editor or browser
2. Ctrl/Cmd + P (Print)
3. Select "Save as PDF" as the destination
4. Adjust margins if needed (use "Fit to Page")

Option 3: Use Online Converter

1. Go to Markdown to PDF
2. Paste the content
3. Click "Convert"
4. Download the PDF

Option 4: Use VS Code (Recommended for Developers)

1. Install the Markdown PDF extension (vscode-pdf)

2. Create a new file called `SAUDI_LEDGER_IMPLEMENTATION.md`
3. Paste the content
4. Right-click → Markdown PDF: Export (pdf)

Option 5: Ask Me to Export

If you're on a system where I can output to a file, ask me to export this as a PDF, and I'll generate it for you directly.